

June 19, 2026

Prof. Mathias Verbeke
KU Leuven
Leuven, Belgium

Dear Prof. Mathias Verbeke,

I am writing to express my strong interest in pursuing a PhD at KU Leuven, under the supervision of your team at the M-Group, focusing on Foundation Models for Generative Design of Mechatronic Products.

As an AI & Data Science Engineer with two engineering degrees, my research interests align perfectly with the intersection of deep learning and mechatronic systems. In my Master's thesis, "Cardiovascular Medical Digital Twin," I engineered Medical_MambaTS, a linear-time state-space model for sub-millisecond bedside inference. This experience taught me the importance of efficient computation and low-latency processing in high-stakes environments.

Furthermore, my work on PI-DeepONet, a Physics-Informed Operator Network using PyTorch Autograd, is highly relevant to your project. This architecture required me to strictly enforce physical fluid constraints (0D Windkessel) within the neural network's loss function. Similarly, in the generative design of mechatronic products, foundation models must rigorously respect physical and mechanical constraints to ensure designs are structurally viable. Additionally, my State Engineer thesis focused directly on a "Gas Turbine Digital Twin," where I modeled thermodynamic degradation and physical anomalies, giving me a deep understanding of mechatronic constraints.

I am confident that my hands-on expertise in constraining deep learning architectures to physical laws makes me an ideal candidate for translating high-level design concepts into buildable mechatronic embodiments. I look forward to discussing my application with you further.

Thank you for considering my application.

Sincerely,
Ismail Aissaoui
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